
Introduction to Operations & Supply Chain Management

Source: Operations and Supply Chain Management, F. Robert Jacobs, McGraw Hill, 17th Edition

1

Introduction

Efficiency at Southwest Airlines

Getting passengers on a plane quickly can greatly affect an airline's cost. Southwest, considered the fastest at turning a plane around, does not assign seats. For Southwest, the goal is to have its airplanes in the air as much as possible. This is difficult, given the multiple short flights that a Southwest jet flies each day.

On average, Southwest's approximately 740 jets fly more than 4,000 flights a day. Without adjusting for jets that are grounded for maintenance, this averages out to over five flights a day per jet. Turning a jet around—from landing to takeoff—is critical to this type of airline, and it has been estimated that Southwest can do this in 30 to 55 minutes, depending on the airport and plane. Think about this: Even at 45 minutes per turn, a Southwest jet still spends about 3.75 hours on the ground each day waiting at the gate. Those precious minutes Southwest can save in loading passengers result in more flights for the airline.



Boarding a Southwest Airlines flight: Quick boarding with no preassigned seats is a key part of the 45-minute turnaround process.

F. Robert Jacobs

I. Introduction—The Elements of OSCM

L01-1: Identify the elements of operations and supply chain management (OSCM).

Really successful firms have a clear and focused idea of how they intend to make money. Whether it be high-end products and services custom tailored to the needs of a single customer or generic and inexpensive commodities bought largely on the basis of cost, competitively producing and distributing these products is a major challenge.

The chapter opening describes the importance of turning planes around quickly at Southwest Airlines. Keeping planes in the air each day is a key factor in the profitability of the company. The process Southwest uses when their planes are on the ground was carefully studied by OSCM (operations and supply chain management) specialists to make it as efficient as possible.

In the context of major business functions, operations and supply chain management involves specialists in product design, purchasing, manufacturing, service operations, logistics, and distribution. These specialists are mixed and matched in many different ways depending on the product or service. For a firm that sells televisions, like Sony, these are the functions responsible for designing televisions, acquiring materials, coordinating equipment resources to convert material to products, moving the product, and exchanging the final product with the customer. Some firms are focused on services, such as hospitals. Here the context involves managing resources, including the operating rooms, labs, and hospital beds used to nurse patients back to health. In this context, acquiring materials, moving patients, and coordinating resource use are keys to success. Other firms are more specialized, such as Amazon. Here purchasing, website services, logistics, and distribution need to be carefully coordinated for success. In our increasingly interconnected and interdependent global economy, the process of delivering finished goods, services, and supplies from one place to another is accomplished by means of mind-boggling technological innovation, clever new applications of old ideas, seemingly magical mathematics, powerful software, and old-fashioned concrete, steel, and muscle.

This book is about doing this at low cost while meeting the requirements of demanding customers. Success involves the clever integration of a great operations-related strategy, processes that can deliver the products and services, and analytics that support the ongoing decisions needed to manage the firm. Our goal in this book is to introduce students to basic operations and supply chain concepts so they understand how things should be done and the importance of these functions to the success of the firm.

No matter what your major is in business, understanding OSCM is critical to your success. If you are interested in the study of finance, you will find that all of the concepts are directly applicable. Just convert all of those widgets to their value in the currency of your choice and you will realize it is all about dollars and cents moving, being stored, and appreciating in value due to exchanges. What you study in finance class is exactly the same, but we look at things in very different ways due to the physical nature of goods and the intangible features of services. The topics presented here are critical to a successful study of marketing, too. If the product or service can't be delivered to the customer at an acceptable cost, then no matter how good your marketing, no one may buy it. Lastly, for those accountants keeping score, operations and supply chain processes generate most of the transactions used to track the financial health of the firm. Understanding why these processes operate the way they do is important to understanding the financial statements of the firm.



1.1 What is Operations and Supply Chain Management?

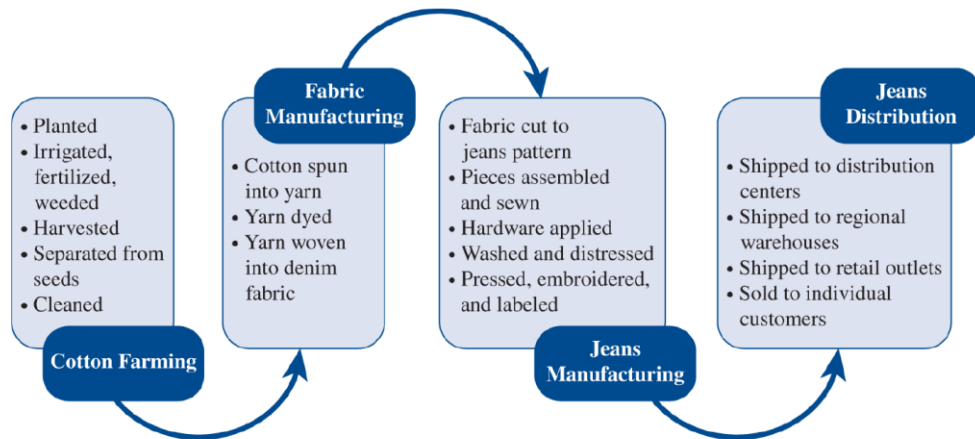
Operations and supply chain management (OSCM) is defined as the design, operation, and improvement of the systems that create and deliver the firm's primary products and services. Like marketing and finance, OSCM is a functional field of business with clear line management responsibilities. OSCM is concerned with the management of the entire system that produces a product or delivers a service. Producing an item such as a pair of jeans, or providing a service such as a cell phone account, involves a complex series of transformation processes.

Exhibit 1.1 shows a supply network for denim jeans sold by a company such as Levi's and Wrangler. Jeans are made from denim, which is a fabric made from cotton, so our supply chain starts with the farmers who grow cotton. In the United States alone, there are over 18,600 cotton farmers. Cotton is used in many products other than jeans. Levi's and Wrangler are two of the largest sellers of denim products in the world.

Cotton farming is a process that typically runs on a yearly cycle with planting starting in the early spring.

EXHIBIT 1.1

Jeans Supply Chain



It takes 6 to 7 months for cotton to grow during which time farmers need to irrigate, fertilize, and weed their crops. The ripe cotton fiber from the plant is then harvested using a mechanical cotton picker. A cotton gin is then used to separate the seeds from the cotton fiber. Finally, the sticky cotton fiber is cleaned and packaged in bales for shipment to cotton markets. Cotton is a globally produced commodity and often sold before it is even produced using auctions with delivery set for dates in the future.

Denim fabric is made by spinning cotton into thin strands of threadlike yarn. The classic denim fabric is blue in color, so some of the yarn is dyed with a blue indigo dye. Four strands of yarn, one white for every three blue, are woven to make the fabric that is then transported to a jeans manufacturing plant.

Manufacturing begins by cutting the fabric to form the jeans pieces. A typical pair of five-pocket jeans is made from 15 pieces of fabric that are assembled and sewn together. The sewn jeans are washed and distressed per the specifications of the design. Buttons, labels, and special embroidering are then applied to each pair of jeans. About 15 minutes of labor is required to make a pair of jeans by hand. The washing, which determines how faded the jeans look, can take from 30 to 360 minutes.

The completed jeans are then shipped to distribution centers, regional warehouses, and finally to retail outlets where they are sold to customers.

Depending on the style and brand of a pair of jeans, the journey through the supply chain can be very different. Take for example, three possible jean supply chains: a luxury–custom pair, a standard pair of men’s jeans, and a low–cost pair of women’s jeans. Assume that all three are sold in retail stores in New York City.

In our example, the luxury–custom pair is sewn in Los Angeles from U.S. cotton denim fabric; these jeans need to get to New York for a fashion photoshoot and an in-store promotion. Cost is not an issue, but time is tight, so they’re made and expedited by overnight airfreight.

The standard pair is made in Mexico in a high-volume factory that has an integrated fabric-producing/jeans manufacturing facility. These jeans are shipped by truck to the New York department store in about a week.

The low-cost fashion pair is sourced through various contract manufacturers in Indonesia, Singapore, Thailand, and Malaysia. Time is less important than protecting slim profit margins. The jeans travel by truck from the manufacturer to a port city, shipped via a container ship to a U.S. West Coast port city, and then shipped by truck or train from the port city to the New York fashion retailer. It takes about six weeks to get an order.

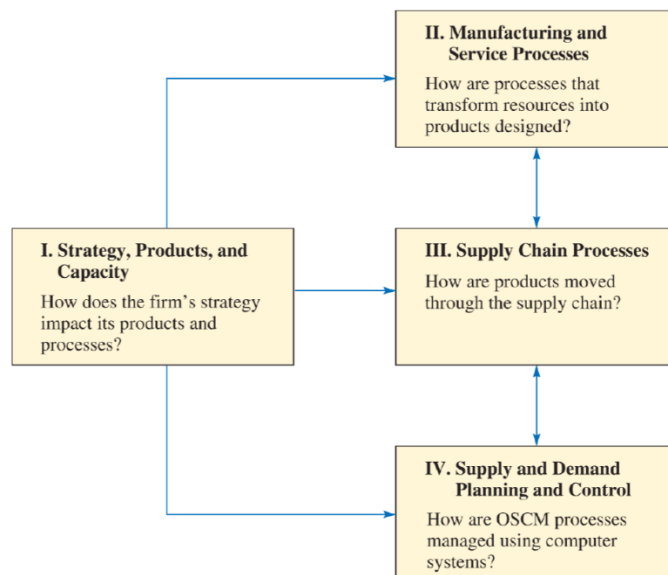


1.2 Distinguishing Operations versus Supply Chain Processes

Success in today’s global markets requires a business strategy that matches the preferences of customers with the realities imposed by complex supply networks. A sustainable strategy that meets the needs of shareholders and employees and preserves the environment is critical.

EXHIBIT 1.2

Questions Answered in Each Section of the OSCM



In the context of our discussion, the terms operations and supply chain take on special meaning. Operations refers to manufacturing and service processes used to transform the resources employed by a firm into products desired by customers. For example, a manufacturing process would produce some type of physical product, such as an automobile or a computer. A service process would produce an intangible product, such as a call center that provides information to customers stranded on the highway

or a hospital that treats accident victims in an emergency room. Planning the use of these processes involves analyzing capacity, labor, and material needs over time. Ensuring quality and making ongoing improvements to these processes are needed to manage these processes.

Supply chain refers to processes that move information and material to and from the manufacturing and service processes of the firm. These include the logistics processes that physically move product and the warehousing and storage processes that position products for quick delivery to the customer. Supply chain in this context refers to providing products and service to plants and warehouses at the input end and the supply of products and service to the customer on the output end of the supply chain.

Planning processes focus on managing and coordinating the use of operations and supply chain resources. Starting with a forecast of demand, resources are planned in increasingly shorter increments of time to match supply inputs with the demand-driven outputs of the firm. These planning activities are completed using integrated computer systems that capture the activities and current status of a firm's resources.

All managers should understand the basic principles that guide the design of transformation processes. This includes understanding how different types of processes are organized, how to determine the capacity of a process, how long it should take a process to make a unit, how the quality of a process is monitored, and how information is used to make decisions related to the design and operation of these processes.

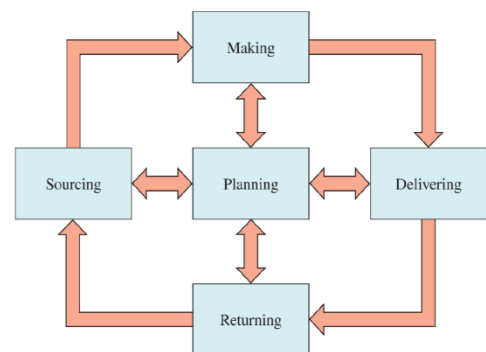
The field of operations and supply chain management is ever-changing due to the dynamic nature of competing in global business and the constant evolution of information technology. So while many of the basic concepts have been around for years, their application in new and innovative ways is exciting. Internet technology has made the sharing of reliable real-time information inexpensive. Capturing information directly from the source through such systems as point-of-sale, radio-frequency identification tags, barcode scanners, and automatic recognition has shifted the focus to understanding both what all the information is saying and also how good the decisions that can be made using it are.

1.3 Categorizing Operations and Supply Chain Processes

Operations and supply chain processes can be conveniently categorized, particularly from the view of a producer of consumer products and services, as planning, sourcing, making, delivering, and returning. Exhibit 1.3 depicts where the processes are used in different parts of a supply chain. The following describes the work involved in each type of process.

**EXHIBIT
1.3**

Operations and Supply Chain Processes



1. **Planning** consists of the processes needed to operate an existing supply chain strategically. Here a firm must determine how anticipated demand will be met with available resources. A major aspect of planning is developing a set of metrics to monitor the supply chain so that it is efficient and delivers high quality and value to customers.

2. **Sourcing** involves the selection of suppliers that will deliver the goods and services needed to create the firm's product. A set of pricing, delivery, and payment processes is needed, along with metrics for monitoring and improving the relationships between partners of the firm. These processes include receiving shipments, verifying them, transferring them to manufacturing facilities, and authorizing supplier payments.

3. **Making** is where the major product is produced or the service provided. The step requires scheduling processes for workers and the coordination of material and other critical resources such as equipment to support producing or providing the service. Metrics that measure speed, quality, and worker productivity are used to monitor these processes.

4. **Delivering** is also referred to as logistics processes. Carriers are picked to move products to warehouses and customers, coordinate and schedule the movement of goods and information through the supply network, develop and operate a network of warehouses, and run the information systems that manage the receipt of orders from customers and the invoicing systems that collect payments from customers.

5. **Returning** involves processes for receiving worn-out, defective, and excess products from customers and support for customers who have problems with delivered products. In the case of services, this may involve all types of follow-up activities required for after-sales support.

KEY IDEA



Companies are positioned in different places in the supply chain. Within the context of their position, they all require planning, sourcing, making, delivering, and returning processes.

To understand the topic, it is important to consider the many different players that need to coordinate work in a typical supply chain. The aforementioned steps of planning, sourcing, making, delivering, and returning are fine for manufacturing and can also be used for the many processes that do not involve the discrete movement and production of parts. In the case of a service firm such as a hospital, for example, supplies are typically delivered on a daily basis from drug and health care suppliers and require coordination between drug companies, local warehouse operations, local delivery services, and hospital receiving. Patients need to be scheduled into the services provided by the hospital, such as operations and blood tests. Other areas, such as the emergency room, need to be staffed to provide service on demand. The orchestration of all of these activities is critical to providing quality service at a reasonable cost.

KEY IDEA



The things produced by a service are intangible. Service processes tend to be highly variable and time dependent compared to goods-producing processes.

1.4 Differences Between Services and Goods

There are five essential differences between services and goods. The first is that a service is

an intangible process that cannot be weighed or measured, whereas a good is a tangible output of a process that has physical dimensions. This distinction has important business implications, because a service innovation, unlike a product innovation, cannot be patented.

Thus, a company with a new concept must expand rapidly before competitors copy its procedures. Service intangibility also presents a problem for customers because, unlike a physical product, customers cannot try it out and test it before purchase.

The second difference is that a service requires some degree of interaction with the customer for it to be a service. The interaction may be brief, but it must exist for the service to be complete. Where face-to-face service is required, the service facility must be designed to handle the customer's presence. Goods, on the other hand, are generally produced in a facility separate from the customer. They can be made according to a production schedule that is efficient for the company.

The third difference is that services, with the big exception of hard technologies (such as ATMs) and information technologies (such as answering machines and automated Internet exchanges) are inherently heterogeneous—they vary from day to day and even hour to hour as a function of the attitudes of the customers and the servers. Thus, even highly scripted work, such as that found in call centers, can produce unpredictable outcomes. Goods, in contrast, can be produced to meet very tight specifications day in and day out with essentially zero variability. In those cases where a defective good is produced, it can be reworked or scrapped.

The fourth difference is that service as a process is perishable and time dependent, and unlike goods, can't be stored. You cannot "come back last week" for an air flight or a day on campus.

And fifth, the specifications of a service are defined and evaluated as a package of design features that affect customer perceptions. Some examples are

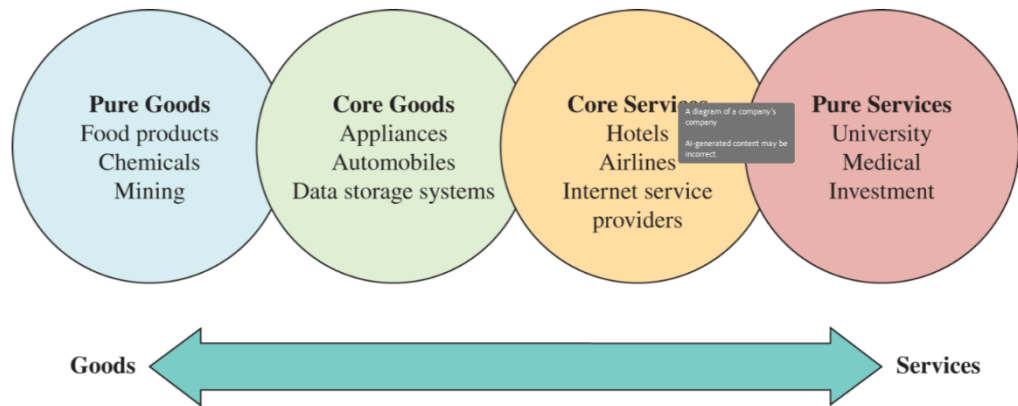
- Supporting facility (location, decoration, layout, architectural appropriateness, supporting equipment)
- Facilitating goods (variety, consistency, quantity of the physical goods that go with the service; for example, the food items that accompany a meal service)
- Explicit services (training of service personnel, consistency of service performance, availability and access to the service, and comprehensiveness of the service)
- Implicit services (attitude of the servers, atmosphere, waiting time, status, privacy and security, and convenience)

Service businesses can be categorized as four main types: (1) businesses that impact human bodies (for example, beauty salons, fitness centers, health clinics); (2) businesses that are directed at physical products (for example, freight transportation, laundry, and environmental landscaping); (3) businesses that are directed at people's minds (for example, advertising, education, arts and entertainment); and (4) businesses directed at risk and money management (for example, insurance, legal services, and banking).

1.5 The Goods-Services Continuum

Almost any product offering is a combination of goods and services. In Exhibit 1.4, we show this arrayed along a continuum of "pure goods" to "pure services." The continuum captures the main focus of the business and spans from firms that just produce products to those that only provide services.

Pure goods industries have become low-margin commodity businesses, and in order to differentiate, they are often adding some services. Some examples are providing help with logistical aspects of stocking items, maintaining extensive information databases, and providing consulting advice.

**EXHIBIT
1.4****The Goods–Services Continuum**

Core goods providers already provide a significant service component as part of their businesses. For example, automobile manufacturers provide extensive spare parts distribution services to support repair centers at dealers. Most of Apple Computer's financial results are driven by sales of the iPhone and other products. Increasingly, the services, which include iTunes and Apple Music, the App Store, iCloud, and Apple Pay, are generating significant revenue. These services provide the glue that sells more Apple products.

Core service providers must integrate tangible goods. For example, your cable television company must provide cable hookup and repair services and high-definition cable boxes. Pure services, such as those offered by a financial consulting firm, may need little in the way of facilitating goods, but what they do use—such as textbooks, professional references, and spreadsheets—are critical to their performance.

1.6. Product–Service Bundling

Product–service bundling refers to a company building service activities into its product offerings for its customers. Such services include maintenance, spare part provisioning, training, and, in some cases, total systems design and research and development (R&D). A well-known pioneer in this area is IBM, which treats its business as a service business and views physical goods as a small part of the “business solutions” it provides its customers. Companies that are most successful in implementing this strategy start by drawing together the service aspects of the business under one roof in order to create a consolidated service organization. The service evolves from a focus on enhancing the product's performance to developing systems and product modifications that support the company's move up the “value stream” into new markets. For example, IBM Cloud has moved the firm to offer an array of new services including analytics, artificial intelligence, Internet of Things, and Blockchain.

This type of strategy might not be the best approach for all product companies, however. A recent study found that while firms that offer product–service bundles generate higher revenues, they tend to generate lower profits as a percent of revenues when compared to focused firms. This is because they are often unable to generate revenues or margins high enough to cover the additional investment needed for service-related costs.

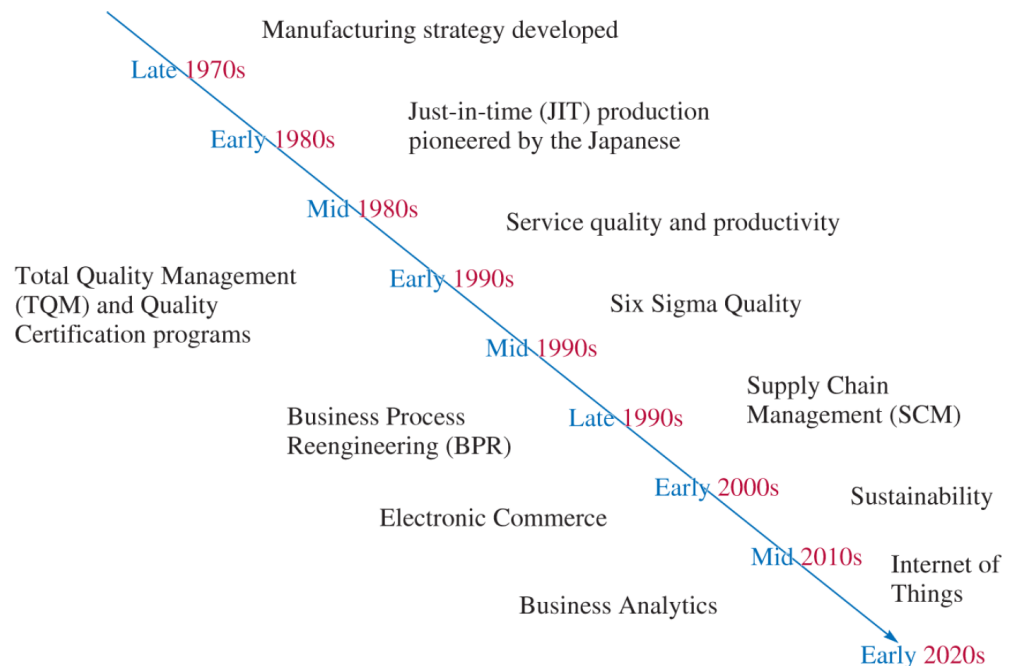
II. The Major Concepts that Define the OSCM Field

LO1-3

Recognize the major concepts that define the operations and supply chain management field.

Our purpose in this section is not to go through all the details of the history of OSCM; that would require us to recount the entire Industrial Revolution. Rather, the focus is on the major operations-related concepts that have been popular since the 1980s. **Exhibit 1.5** will help clarify the dates as you read about the concepts. Where appropriate, how a supposedly new idea relates to an older idea is discussed. (We seem to keep rediscovering the past.)

EXHIBIT 1.5 Timeline Depicting When Major OSCM Concepts Became Popular



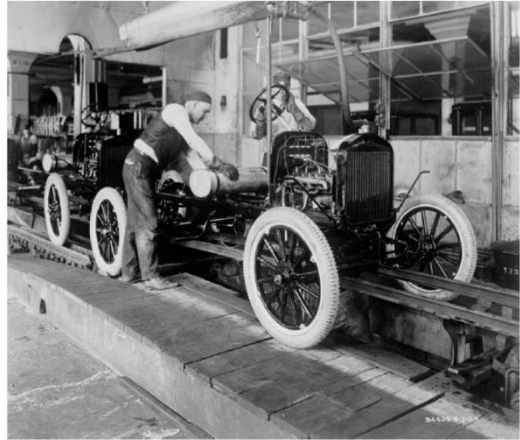
Source: F. Robert Jacobs

Manufacturing Strategy Paradigm The late 1970s and early 1980s saw the development of the **manufacturing strategy** paradigm, which emphasized how manufacturing executives could use their factories' capabilities as strategic competitive weapons. Central to this thinking was the notion of manufacturing trade-offs among such performance measures as low cost, high quality, and high flexibility.

Lean Manufacturing, JIT, and TQC The 1980s saw a revolution in the management philosophies and technologies by which production is carried out. **Just-in-time (JIT)** production was the major breakthrough in manufacturing philosophy. Pioneered by the Japanese, JIT is an integrated set of

activities designed to achieve high-volume production using minimal inventories of parts that arrive exactly when they are needed. The philosophy—coupled with **total quality control (TQC)**, which aggressively seeks to eliminate causes of production defects—is now a cornerstone in many manufacturers' production practices, and the term **lean manufacturing** is used to refer to the set of concepts.

Of course, the Japanese were not the first to create a highly integrated, efficient production system. In 1913, Henry Ford developed an assembly line to make the Model-T automobile. Ford developed a system for making the Model-T that was constrained only by the capabilities of the workforce and existing technology. Quality was a critical prerequisite for Ford: The line could not run steadily at speed without consistently good components. On-time delivery was also critical for Ford; the desire to keep workers and machines busy with materials flowing constantly made scheduling critical. Product, processes, materials, logistics, and people were well integrated and balanced in the design and operation of the plant.



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Service Quality and Productivity The unique approach to quality and productivity pioneered by McDonald's has been so successful that it stands as a reference point in thinking about how to deliver high-volume standardized services.

Total Quality Management and Quality Certification Another major development was the focus on **total quality management (TQM)** in the late 1980s and 1990s. Helping the quality movement along was the Baldrige National Quality Award, started in 1987 under the direction of the National Institute of Standards and Technology. The Baldrige Award recognizes companies each year for outstanding quality management systems.

The ISO 9000 certification standards, created by the International Organization for Standardization, now play a major role in setting quality standards for global manufacturers. Many companies require that their vendors meet these standards as a condition for obtaining contracts.

Business Process Reengineering The need to become lean to remain competitive in the global economic recession in the 1990s pushed companies to seek innovations in the processes by which they run their operations. The **business process reengineering (BPR)** approach seeks to make revolutionary changes as opposed to evolutionary changes (which are commonly advocated in TQM). It does this by taking a fresh look at what the organization is trying to do in all its business processes, and then eliminating non-value-added steps and computerizing the remaining ones to achieve the desired outcome.

Six Sigma Quality Originally developed in the 1980s as part of total quality management, **Six Sigma** in the 1990s saw a dramatic expansion as an extensive set of diagnostic tools was developed. These tools have been taught to managers as part of "Green and Black Belt Programs" at many corporations. The tools are now applied not only to the well-known manufacturing applications, but also to non-manufacturing processes such as accounts receivable, sales, and R&D. Six Sigma has been applied to environmental, health, and safety services at companies and is now being utilized in R&D, finance, information systems, legal, marketing, public affairs, and human resource processes.

Supply Chain Management The central idea of supply chain management is to apply a total system approach to managing the flow of information, materials, and services from raw material suppliers through factories and warehouses to the end customer. Trends such as outsourcing and **mass**

customization are forcing companies to find flexible ways to meet customer demand. The focus is on optimizing core activities to maximize the speed of response to changes in customer expectations.

Electronic Commerce The quick adoption of the Internet and the World Wide Web during the late 1990s was remarkable. The term **electronic commerce** refers to the use of the Internet as an essential element of business activity. The use of web pages, forms, and interactive search engines has changed the way people collect information, shop, and communicate. Companies such as Amazon and Alibaba have created a new type of online shopping experience for busy consumers.

Sustainability and the Triple Bottom Line Sustainability is the ability to maintain balance in a system. Management must now consider the mandates related to the ongoing economic, employee, and environmental viability of the firm (the **triple bottom line**). Economically, the firm must be profitable. Employee job security, positive working conditions, and development opportunities are essential. The need for non-polluting and non-resource-depleting products and processes presents new challenges to operations and supply managers.

Business Analytics involves the analysis of data to better solve business problems. Not that this is something new: Data have always been used to solve business problems. What is new is the reality that so much more data are now captured and available for decision-making analysis than were available in the past. In addition, mathematical tools are now readily available that can be used to support the decision-making process.

In the past, most analysis involved the generation of standard and ad hoc reports that summarized the current state of the firm. Software allowed querying and “drill down” analysis to the level of the individual transaction, useful features for understanding what happened in the past. Decision making was typically left to the decision maker based on judgment or simple alerting rules. The new “analytics” movement takes this to a new level using statistical analysis, forecasting to extrapolate what to expect in the future, and even optimization, possibly in real time, to support decisions. These mathematical results can be used either to support the decision maker or to automate decision making. The **Internet of Things** refers to the billions of devices that are connected to the Internet.

Data coming from these devices open new opportunities such as the tracking of assets, intelligent monitoring of equipment to prevent failures, and dynamic reaction to changes in demand.

Current Issues in Operations and Supply Chain Management

OSCM is a dynamic field, and issues arising in global enterprise present exciting new challenges for OSCM managers. Looking forward to the future, the following are major challenges that will face managers in the future:

1. Disruptions in global supply chains. The COVID-19 pandemic has shown how difficult syncing diverse “players” in the supply chain can be. The major regions, especially North America, the European Union, and China, struggled to recover from the pandemic. The diverse economies recovered at different rates, resulting in constantly changing bottlenecks that strangled the economies of the regions. The changes in work patterns, sales platforms, distribution, and value proposition have been profound. While these changes have resulted in many new opportunities, they have also resulted in new challenges.

2. Uncertainty in global tariffs and regulations. Global companies face significant uncertainty in how tariffs will be assessed in major regions, especially in North America, the European Union, and China. This uncertainty makes long-term planning related to the location of manufacturing plants and the selection of suppliers difficult. Until a new global balance emerges, OSCM managers are forced to think short term, which makes being efficient difficult.

3. Difficulty in hiring and keeping employees. The void in manufacturing, service, and logistics workers is a critical problem. There is a supply chain talent emergency with shortages in executives,

managers, analysts, and even truck drivers. Management-level workers in the field need a set of skills that range from being technically savvy to being articulate with great communication skills. People with these skills are unique and highly sought after. The opportunities are great for those up to the challenge.

4. **Adaptability to change in business technology and infrastructure.** The challenge of changing the business to take advantage of technology and methods for doing business is essential to future success. OSCM is faced with a constant barrage of change. A few examples include the new Internet platforms for doing business as used by Amazon, product technologies driven by connected interfaces as offered by Alphabet (Google), and transportation technologies such as electric trucks offered by Tesla. Failure to adapt results in a firm not being able to compete quickly due to obsolete products and inefficient processes.

III. Efficiency, Effectiveness, and Value

LO1-4

Evaluate the efficiency of a firm.

Compared with most of the other ways managers try to stimulate growth—technology investments, acquisitions, and major market campaigns, for example—innovations in operations are relatively reliable and low cost. As a business student, you are perfectly positioned to come up with innovative operations-related ideas. You understand the big picture of all the processes that generate the costs and support the cash flow essential to the firm's long-term viability.

Through this book, you will become aware of the concepts and tools now being employed by companies around the world as they craft efficient and effective operations. **Efficiency** means doing something at the lowest possible cost. Later in the book, we define this more thoroughly, but roughly speaking, the goal of an efficient process is to produce a good or provide a service by using the smallest input of resources. In general, these resources are the material, labor, equipment, and facilities used in the OSCM processes.

Effectiveness means doing the right things to create the most value for the company. For example, to be effective at a grocery store, it is important to have plenty of operating checkout lines even though they may often stand idle. This is a recognition that customers' time is valuable and that they do not like waiting to be served in the checkout line. Often, *maximizing effectiveness and efficiency at the same time creates conflict between the two goals*. We see this trade-off every day in our lives. At the checkout lines, being efficient means using the fewest people possible to ring up customers. Being effective, though, means minimizing the amount of time customers need to wait in line.

Related to efficiency and effectiveness is the concept of **value**, which can be abstractly defined as quality divided by price. Here, quality is the attractiveness of the product, considering its features and durability. If you can provide the customer with a better car without changing price, value has gone up. If you can give the customer a better car at a *lower* price, value goes way up. A major objective of this book is to show how smart management can achieve high levels of value.